order to find the right clues to the concept of causality, we must go still a little deeper.

In the case of weather forecasts, the thought suggests itself that their unreliability is due merely to the size and complicated nature of the object of the analysis, i.e. the atmosphere. If we single out a small portion of it—for instance, a cubic foot of air—we will be far more likely to make accurate predictions about its reaction to external influences, such as compression, heat, moisture, etc. We are familiar with certain laws of physics which enable us to predict, with more or less certainty, the readings of the corresponding measurements, such as the increase in pressure, rise in temperature, condensation, etc.

However, if we observe things still a little more closely, we shall soon reach a very remarkable conclusion. Simple as we may make the conditions, and precise as our measuring instruments may be, we shall never succeed in calculating in advance the results of the actual measurements with an absolute accuracy, in other words in making the predicted value of a magnitude agree to the last decimal place with the figure actually registered by the instruments; there always remains a certain margin of uncertainty—in contrast to the calculations in pure mathematics, as in the case of the square root of 2 or of π, which can be given accurately to any number of decimal places. And whatever applies to mechanical and thermal processes, holds true for all fields of physics, including electricity and optics.

Thus, all the above cited experiences force us to recognize the following principle as a firmly established fact: *It is never possible to predict a physical occurrence with unlimited precision.* If we now compare this principle with the proposition accepted above as our starting point, namely that an occurrence is causally determined if it can be predicted with certainty, we find ourselves facing an unavoidable dilemma: We may elect *either* to adhere literally to the exact wording of our basic proposition, in which case there cannot exist even one single instance in nature where a causal relationship would have to be assumed to prevail—*or* to subject that basic proposition to a certain modification, so designed as to provide room for the presupposition of strict causality.

A number of contemporary physicists and philosophers have chosen the first alternative. I shall refer to them here as *indeterminists*. They claim that genuine causality, strict regularity, actually does not exist in nature but is merely an illusion created by the operation of certain rules which are never of an exact universal validity, even though they often come very near to it. Upon closer consideration, the indeterminist discovers a statistical root in every law of physics, including the law of gravity and of electrical attraction; he regards them, one and all, as laws of probability, relating only to mean values of many similar observations, possessing only an approximate validity for individual